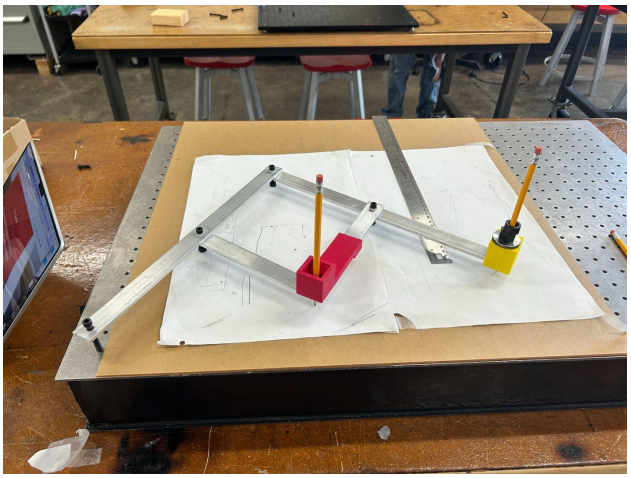
KASIN ITO-SATO Project Portfolio | Mechanical Engineering, Loyola Marymount University, May 2027
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2.0x Pantograph: Design, Tolerancing, and Manufacturing

Design and Manufacturing (MECH 2100), LMU | Team of two, with Evan Soleimani | Fall 2024

Goal

Design and build a pantograph that reproduces an image at exactly 2.0x, mounts to an optical breadboard, uses at least two materials and three manufacturing processes, and holds a replaceable marker.



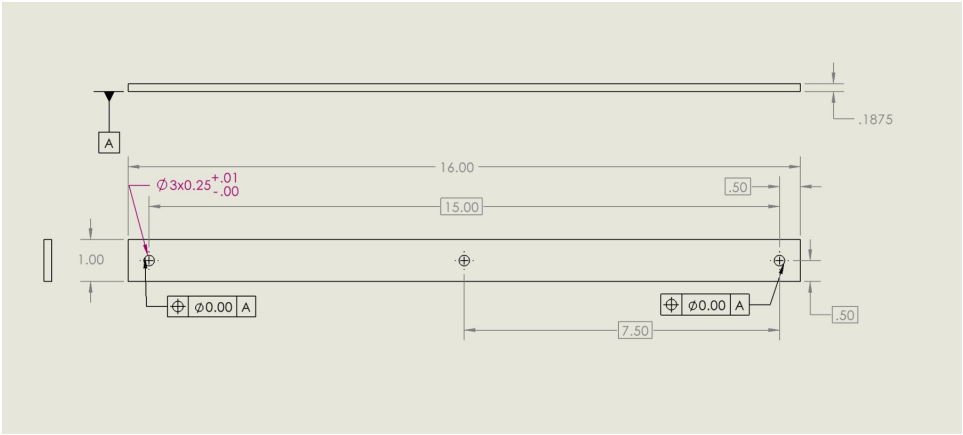
Finished pantograph mounted to the optical breadboard.

Design and build

- Aluminum bar linkage with 3D-printed marker and tracer holders.
- Three processes: mill, horizontal bandsaw, and 3D printing.
- Built the SolidWorks model as the blueprint for the hardware, then used the mill's digital readout to locate hole spacing.

Tolerancing (GD&T)

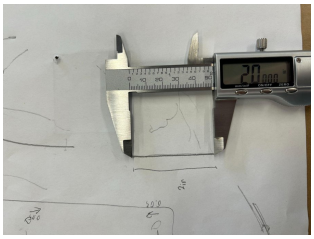
Detailed each link with true-position callouts at maximum material condition for floating 1/4-20 fasteners. Holes were specified at Ø0.25 in +0.01/−0.00, giving a hole MMC of 0.25 in against a fastener MMC of 0.25 in (T = H − F = 0).



Link drawing with Ø0.25 in hole tolerance and true-position callouts referenced to datum A.

Results

- 2 in test line reproduced at exactly 4.0 in (2.00x).
- 2 in x 2 in box reproduced between 1.98x and 2.00x, never above 2.00x.



Original: 2.0 in



Reproduction: 4.0 in

What I would change

- Lock hole sizes before machining. A first hole size did not fit, and re-drilling added error to the linkage.
- Design the marker holder for one utensil. Making it fit many pens cost precision.
- As-built parts matched the CAD model closely; small bandsaw length variations had little effect because hole spacing was set on the mill.

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SAE Aero Design Capstone (Regular Class 2027)

LMU senior capstone | Five-person team | 2026 to 2027, in progress

Designing a multi-motor electric RC aircraft that carries 2-liter bottle payloads under the 2027 SAE rules: 55 lb maximum gross weight, 120 in maximum length, 72 to 96 in wingspan, two or four electric motors, and a 2200 mAh battery limit. I work across aero, structures, propulsion, and controls.

- Compared two- and four-motor layouts and landing-gear configurations against the rule limits.
- Laid out wiring and connections integrating the motors, ESCs, and battery into the airframe.
- Sized the cargo bay around bottle dimensions and payload capacity.
- Built the team's project-management workbook for tasks, phases, and schedule.

Self-Initiated HVAC Calculation Tools

WSP, Mechanical Engineering Intern | 2025 to 2026

Exhaust airflow, building pressurization, and demand-controlled ventilation calculations came up on project after project and were slow to do by hand. On my own initiative, I researched the equations and code requirements and built three Excel tools that return consistent results from one set of inputs. The team first questioned them because they used newer data than the team usually worked from, then adopted them.

Heat Sink and Forced-Convection Testing

Heat Transfer Lab (MECH 3230), LMU | Spring 2026

Built an Arduino UNO data-acquisition setup with three MAX6675 thermocouples to log steady-state temperatures of a cartridge-heated block, then reduced the data to convection coefficients: about 13 W/m²K with a fin array in natural convection and 48 W/m²K with forced air. A second test varied fan speed:

Air speed (m/s)	Power (W)	Ambient (°C)	Steady state (°C)	R = ΔT / Q (K/W)
4.3	10	21	57.25	3.62
5.8	12	21	56.75	2.98
6.2	14.06	21	61.75	2.90

Thermal resistance fell from about 3.6 to 2.9 K/W as air speed rose from 4.3 to 6.2 m/s.

Motor Position Control and Motion Sensing

Dynamics and Control I & II, LMU | 2025 to 2026

- Wrote interrupt-based Arduino code to read a quadrature encoder (90 x 64 PPR), drive a motor forward and back from push buttons, and display position on an OLED, then plotted position and velocity over time.
- Calibrated a two-axis accelerometer to separate tangential and normal acceleration.
- Tuned pitch and x-axis controllers in MATLAB/Simulink with root-locus methods for a two-axis tracking task, comparing PI and PID control.